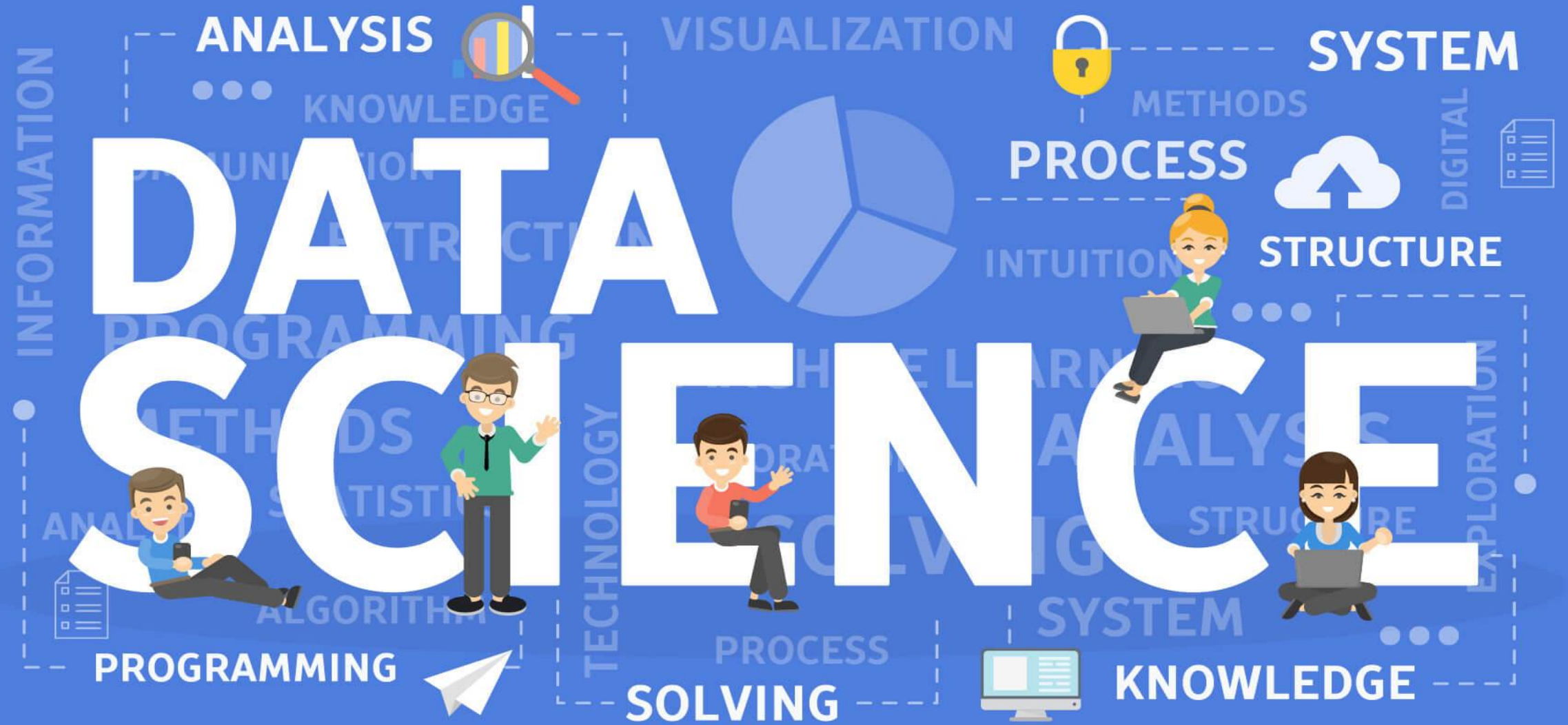




Ex:No.#04

Read Files and Explore various Commands

File Operations

To read data from the text files, excel and the web and exploring various commands for doing descriptive analytics on the iris data set

Software Requirements

- ❖ Python 3.x
- ❖ Jupyter Notebook / Google Colab
- ❖ Pandas - NumPy - Matplotlib - Seaborn - OpenPyXL (for Excel files)

Task	Command
Read text file	<code>pd.read_csv('file.txt', sep=',') or sep='\t'</code>
Read Excel file	<code>pd.read_excel('file.xlsx', sheet_name=...)</code>
Read from web	<code>pd.read_csv(url)</code>
Structure	<code>.shape, .dtypes, .info(), .head(), .tail()</code>
Missing/duplicates	<code>.isnull().sum(), .duplicated().sum()</code>
Summary stats	<code>.describe(), .mean(), .median(), .std(), .var()</code>
Categorical summary	<code>.value_counts(), .unique()</code>
Grouped stats	<code>.groupby('species').describe()</code>
Relationships	<code>.corr(), sns.heatmap(), sns.pairplot()</code>
Distribution shape	<code>.skew(), .kurt(), .hist()</code>

Steps	Description
1. Install	❖ pip install pandas numpy matplotlib seaborn openpyxl

Package	Purpose
pip	Python's package manager used to install and manage external libraries.
pandas	Used for reading, cleaning, manipulating, and analyzing data in tables (DataFrames). It supports CSV, Excel, SQL, JSON, and more.
numpy	Provides support for numerical computations, arrays, matrices, and mathematical operations. It is the foundation for many data science libraries.
matplotlib	Used to create basic data visualizations such as line charts, bar charts, scatter plots, histograms, and pie charts.
seaborn	Built on Matplotlib, it provides advanced statistical visualizations with attractive default styles. It also includes sample datasets like the Iris dataset.
openpyxl	Enables Pandas to read from and write to Microsoft Excel (.xlsx) files.

Basic Function in Pandas

- ❖ read_csv()
- ❖ read_excel()
- ❖ head()
- ❖ tail()
- ❖ info()
- ❖ describe()
- ❖ shape
- ❖ columns
- ❖ dtypes
- ❖ value_counts()
- ❖ mean()
- ❖ median()
- ❖ mode()
- ❖ std()
- ❖ min()
- ❖ max()

Steps	Description
2. Read from web	<pre>import pandas as pd import numpy as np url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv" iris = pd.read_csv(url) print(iris.head())</pre>
3. Save as Text File	<pre>iris.to_csv("iris.txt", index=False)</pre>
4. Save as Excel File	<pre>iris.to_excel("iris.xlsx", index=False)</pre>
Error	If openpyXL module not found install in another cell as <pre>%pip install openpyxl</pre>

Steps	Description
5. Read from Text file	<code>pandas.read_csv()</code> is the general-purpose command for delimited text files. The delimiter is controlled with the <code>sep</code> argument (, for CSV-style text, \t for tab-separated
Code	- One Python script — every example prints real output - Each section ends with a challenge; solutions live at the bottom - Covers creation, filtering, missing data, groupby, merges, reshaping, apply/map, time series # Comma-delimited .txt file <pre>df_txt = pd.read_csv('iris.txt') print(df_txt.shape) df_txt.head()</pre> Attached separately as pandas_examples_and_challenges.py
Code	# Tab-delimited .txt file -- same command, different separator <pre>df_txt_tab = pd.read_csv('iris_tab.txt', sep='\t') df_txt_tab.head()</pre>

Steps	Description
5. Read from Text file	# Reading a text file with the plain built-in file handle (no pandas), # useful when you need line-by-line control rather than a DataFrame
Code	<pre> with open('iris.txt') as f: header = f.readline().strip() first_data_line = f.readline().strip() print('Header:', header) print('First row:', first_data_line) </pre>

Steps	Description
6. Read from Excel file	pandas.read_excel() reads .xlsx/.xls files. It needs the openpyxl engine for .xlsx files (installed automatically with pandas in most environments; install with pip install openpyxl if you get an engine error). • One Python script — every example prints real output • Each section ends with a challenge; solutions live at the bottom • Covers creation, filtering, missing data, groupby, merges, reshaping, apply/map, etc Use sheet_name to pick a specific sheet, or sheet_name=None to read every sheet into a dictionary of DataFrames.
Code	<pre>df_excel = pd.read_excel('iris.xlsx', sheet_name='iris')df_excel.head()</pre> • Attached separately as pandas_examples_and_challenges.py

Steps	Descriptive analytics on the Iris dataset
7. Basic Structure	df.shape # (rows, columns) df.dtypes # data type of each column df.info() # combined structure summary df.head() # first 5 rows df.tail(3) # last 3 rows df.sample(5, random_state=1) # random sample of rows
8. Missing values and duplicates	df.isnull().sum() # missing values per column df.duplicated().sum() # count of duplicate rows
9. Summary Statistical	df.describe() # count, mean, std, min, quartiles, max for numeric columns df.describe(include='all') # include categorical (species) column too df['species'].value_counts() # class balance df['species'].unique() # distinct species labels

Steps	Descriptive analytics on the Iris dataset
10. Grouped descriptive statistics (by species)	<pre>df.groupby('species').mean(numeric_only=True) df.groupby('species').describe().T # full stats broken out per species df.groupby('species').agg(['mean', 'std', 'min', 'max'])</pre>
11. Relationships between variables	<pre>df.corr(numeric_only=True) # correlation matrix sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm') plt.title('Correlation matrix - Iris numeric features') plt.show()</pre>
	<pre>df.nlargest(5, 'sepal_length') # top 5 rows by a column df.nsmallest(5, 'petal_width') # bottom 5 rows by a column</pre>

Steps	Descriptive analytics on the Iris dataset
12. Visual descriptive analytics	<pre>df.hist(figsize=(9, 6)) plt.suptitle('Distribution of each feature') plt.tight_layout()plt.show()</pre> One Python script — every example prints real output
	<pre>sns.boxplot(data=df, x='species', y='petal_length') plt.title('Petal length by species') plt.show()</pre> Attached separately as pandas_examples_and_challenges.py
	<pre>sns.pairplot(df, hue='species') plt.show()</pre>

Common Data Frames Functions



Function	Purpose
head()	Display first 5 rows
tail()	Display last 5 rows
info()	DataFrame information
describe()	Statistical summary
shape	Number of rows and columns
columns	List of column names
loc[]	Access rows by label
iloc[]	Access rows by index
drop()	Remove rows/columns
rename()	Rename columns
sort_values()	Sort data
isnull()	Detect missing values
fillna()	Fill missing values
dropna()	Remove missing values
to_csv()	Save to CSV
read_csv()	Read CSV file